RFI DETECTION IN SENTINEL-1 SAR IMAGERY VIA PSEUDO-LABEL DISTILLATION AND MULTI-RESOLUTION ENSEMBLE

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ABSTRACT

We present a detection pipeline for Radio Frequency Interference (RFI) in Sentinel-1 Synthetic Aperture Radar (SAR) quicklook imagery, achieving 0.4776 $\text{mAP}_{50\text{-}95}$ on the held-out test set of the ESA ClearSAR Track 1 challenge. RFI manifests as thin horizontal stripes (median height $\approx$10 px, median width $\approx$140 px) in variable-size Sentinel-1 RGB quicklooks (median $\approx 515\times342$ px), a small-object geometry that differs markedly from standard COCO categories. The pipeline combines (1) a YOLO26x base detector with SAR-specific augmentation and pixel-space Normalised Wasserstein Distance (NWD) regression loss; (2) pseudo-label distillation from a multi-fold ensemble teacher onto unlabelled test imagery; (3) multi-resolution test-time augmentation (TTA) fused via Weighted Box Fusion (WBF); and (4) selective cross-fold ensembling that discards the domain-overlapping fold before final fusion. On a 401-image leak-isolated meta-validation set, the distilled student source gains $+0.077$ $\text{mAP}_{50\text{-}95}$ over a cold per-fold baseline (pseudo-distillation $+0.040$, multi-resolution TTA $+0.035$, fold dropping $+0.002$). On the held-out test set this source is redundant with the ensemble teacher: the full distillation–TTA–blend stack adds only $+0.0021$ over submitting the teacher alone, and $+0.0056$ over a plain 5-fold ensemble. We report both measurements; the gap between them is the paper's main finding.

***Index Terms*—** Ensemble, pseudo-label distillation, radio-frequency interference, SAR imagery, small object detection.

1. INTRODUCTION

Synthetic Aperture Radar (SAR) sensors aboard Sentinel-1 provide continuous global monitoring at C-band (5.4 GHz). As terrestrial wireless infrastructure has expanded, Radio Frequency Interference from communication networks and other emitters has become an increasingly common data quality problem [1]. RFI corrupts focused SAR imagery as bright horizontal azimuth lines and, in quicklook products, as rectangular stripes spanning the full image width at affected azimuth positions.

The ESA ClearSAR Track 1 challenge [2] frames RFI detection as a standard object detection task: given a variable-size (median $\approx 515\times342$ px) 8-bit RGB quicklook, localise all RFI-contaminated azimuth lines with axis-aligned bounding boxes and report results in COCO format [3]. The metric is COCO mAP averaged over IoU thresholds 0.50–0.95 ($\text{mAP}_{50\text{-}95}$).

RFI stripes are geometrically unlike any COCO category: a median per-box aspect ratio of $\approx$8:1 (median height $\approx$10 px, median width $\approx$140 px), with box heights spanning roughly 4–120 px (5th–95th percentile) and widths up to $\approx$500 px. This geometry creates several detection challenges:

- IoU-based regression loss provides no gradient when predicted and target boxes do not overlap — common when height predictions are off by only a few pixels.
- SAR side-looking geometry breaks horizontal symmetry: left–right reflections change the physically meaningful look direction.
- Multi-scale test-time augmentation above 640 px amplifies speckle noise relative to the signal, hurting recall at larger resolutions.

We make three contributions:

- We adapt a YOLO26x detector to RFI stripe geometry with a pixel-space NWD regression loss and SAR-aware augmentation, recovering a usable gradient for thin, non-overlapping boxes (Section 3).
- We assemble an ensemble-teacher pseudo-label distillation, multi-resolution TTA and selective cross-fold fusion pipeline that reaches 0.4776 $\text{mAP}_{50\text{-}95}$ (28th of 172) on the held-out test set (Section 3).
- We evaluate each component on a leak-isolated meta-validation set and on the held-out test set, showing the distilled source gains $+0.077$ $\text{mAP}_{50\text{-}95}$ in isolation but only $+0.0021$ over the teacher on unseen data (Section 4).

2. RELATED WORK

SAR object detection. Deep-learning SAR detection has been studied extensively for ships [4, 5] and aircraft [6]. These targets are compact (aspect ratios $\approx$1–4:1); RFI stripes are qualitatively different. The nearest analogous tasks are lane-line and wire/cable detection in RGB imagery, both of which favour architectures with elongated receptive fields.

Small object detection. Standard detectors struggle when objects occupy fewer than 32×32 pixels owing to limited feature map resolution and gradient imbalance [7]. Proposals for improvement include feature pyramid networks [8], transformer-based decoders [9], and modified regression losses. Normalised Wasserstein Distance (NWD) [10] replaces IoU regression with a Wasserstein distance between Gaussian distributions centred on predicted and target boxes, yielding non-zero gradients for non-overlapping small targets.

Pseudo-label distillation. Knowledge distillation via pseudo-labels [11, 12] trains a student on predictions from a stronger teacher, transferring soft knowledge encoded in the teacher's confidence scores. When the teacher is a multi-fold ensemble and the student is fine-tuned on unlabelled in-distribution data, the approach is sometimes called noisy student training [12] or self-training [13].

Test-time augmentation and WBF. Ensemble methods aggregate predictions across multiple checkpoints or image transformations. WBF [14] clusters overlapping detections and computes a weighted average of box coordinates, avoiding the precision-recall trade-off of NMS-based fusion, and is widely used when combining predictions at multiple resolutions.

3. METHOD

The pipeline has four stages: base training, pseudo-label generation, fine-tuning with distillation, and multi-resolution TTA ensemble. It is summarised in Figure 1.

3.1. Data preparation

Dataset split. The 3,154 labelled images are divided into five stratified folds. We reserve a *meta-validation set* of 401 images sampled from the fold-0 validation pool (random seed 42) for leak-isolated local evaluation. All 2,753 remaining images form the meta-training pool.

SAR-specific augmentation. SAR side-looking geometry imposes that left–right reflections change the physical meaning of the image (near-range versus far-range swaps), while vertical (azimuth) flips change the look direction. We therefore set `fliplr` $= 0$ and `flipud` $= 0$. Ablation confirmed that enabling horizontal flip degrades $\mathrm{mAP}_{50\text{-}95}$ by 6.5% relative. Geometric augmentations are restricted to conservative translations ($\pm 5\%$ of image size) and mild scale perturbations ($\pm 25\%$), preventing thin RFI boxes from shrinking below reliable detection thresholds. Colour jitter (HSV) is suppressed because SAR quicklooks carry no chromatic information. Base training additionally uses an offline SAR-aware copy-paste augmentation (one augmented copy per image, three RFI pastes per copy, max paste IoU 0.05); an ablation showed it gives no gain on top of pseudo-distillation, so it is disabled during fine-tuning.

Contrast enhancement (CLAHE). As an offline preprocessing step we apply Contrast-Limited Adaptive Histogram Equalisation (CLAHE, clip limit 3.0, 8×8 tile grid) to the luminance (L) channel in LAB colour space, sharpening thin low-contrast stripes against speckle without amplifying the global background.

3.2. Base detector: YOLO26x with NWD loss

We use YOLO26x [15] ($\approx$59 M parameters) pre-trained on COCO as the base detector, without the optional P2 detection head: YOLO26's built-in small-target modules (STAL and progressive loss) already address small objects, and an ablation showed the added P2 head plateaued well below baseline on this data. The risk of overfitting the 2,753-image training set is addressed by the pseudo-label distillation stage (Section 3.3) rather than by shrinking the backbone. The detector layout is summarised in Figure 2.

NWD regression loss. Standard YOLO regression uses Complete IoU (CIoU) [16] loss. For thin stripes where predicted and target boxes frequently do not overlap vertically, CIoU is identically zero, providing no gradient. We replace it with a blend of CIoU and pixel-space NWD,

$$\mathcal{L}_{\text{box}} = (1-\alpha)\,\mathcal{L}_{\text{CIoU}} + \alpha\,\mathcal{L}_{\text{NWD}}, \qquad \alpha = 0.5. \tag{1}$$

Following [10], each box is modelled as a 2-D Gaussian $p(b) = \mathcal{N}(\mu_b, \Sigma_b)$ with mean $\mu_b = (c_x, c_y)^\top$ and diagonal covariance $\Sigma_b = \mathrm{diag}(w^2/4,\ h^2/4)$, and NWD is defined via the closed-form Wasserstein-2 distance,

$$\mathcal{L}_{\text{NWD}}(b, \hat{b}) = 1 - \exp\left(-\frac{\sqrt{W_2^2(p_b, p_{\hat{b}})}}{C}\right), \tag{2}$$

where $C = 12.8$ pixels is a normalisation constant. An implementation detail to note is that ultralytics evaluates the box loss on coordinates normalised to the network input size, not in pixels. We rescale the positive boxes to pixel units before computing NWD so that C is interpreted in pixel space; computing NWD on the normalised coordinates directly leaves C far too large, saturates the similarity to a near-constant, and removes the gradient in the small-height regime characteristic of RFI stripes (median $\approx$10 px).

Hyperparameters. Base training runs for up to 60 epochs (early stopping, patience 15) with the MuSGD optimiser, initial learning rate $\eta_0{=}10^{-3}$ decaying linearly to $\eta_f{=}10^{-5}$, 3 warmup epochs, batch 32, 640 px resolution, box-loss weight 12.0.

3.3. Pseudo-label distillation

Teacher ensemble. The teacher (denoted V12) is our strongest pre-distillation ensemble: a single-stage WBF over multiple YOLO26x fold and multi-resolution-TTA sources (with the fold-2 source replaced by a CLAHE-retrained variant) together with one RF-DETR-Large [17] fold checkpoint weighted $1.5\times$,

$$\hat{D}_{\text{teacher}} = \mathrm{WBF}(\{D_f\}_f,\ D_{\text{RF}};\ \tau{=}0.70,\ \texttt{conf\_type=max}), \tag{3}$$

where τ is the IoU clustering threshold. V12 scores 0.4755 on the held-out test set. WBF parameters (τ=0.70, `conf_type=max`) were fixed after sweeping $\tau \in \{0.65, 0.70, 0.75\}$ and four confidence aggregation modes; all variants outside (0.70, `max`) degraded performance. The marginal value of the RF-DETR component is ablated in Section 4.

Pseudo-labelling the test set. The teacher is applied to the 786 unlabelled test images with a confidence threshold of 0.75, keeping at most 20 boxes per image, producing 600 pseudo-labelled images after discarding low-density detections. These are merged with each fold's training split to form *fold-augmented* datasets.

Student fine-tuning. For each fold $f \in \{1, 2, 4\}$, a student is fine-tuned from the per-fold base checkpoint of Section 3.2 on the fold-augmented dataset, with the backbone frozen for the first 10 epochs (`freeze=10`), AdamW [18] $\eta_0{=}10^{-4}$, 15 total epochs, batch 24, and the same NWD loss blend as base training. Fold 0 is omitted because its base partition overlaps the meta-validation pool: the fold-0 fine-tuned student produces correlated rather than complementary predictions, and including it empirically degrades ensemble accuracy (Section 4).

3.4. Multi-resolution TTA and final fusion

Each fine-tuned student is applied to the meta-validation and test images at four resolutions $\mathcal{I} = \{608, 640, 672, 704\}$ pixels. Predictions across the four resolutions are fused per-fold via WBF (IoU=0.70, `conf_type=max`), producing one TTA-fused prediction set per fold. Resolutions above 704 px were tested and found to degrade performance due to amplified speckle aliasing at YOLO's feature pyramid strides. Flip-based TTA (horizontal/vertical) is explicitly disabled.

Per-fold TTA predictions are then fused into a single super-source S^* via WBF with equal weights,

$$S^* = \mathrm{WBF}\Big(D_1^{\text{TTA}}, D_2^{\text{TTA}}, D_4^{\text{TTA}};\ w{=}1.0,\ \tau{=}0.70\Big). \tag{4}$$

The final submission candidate (denoted V17) is a two-source blend of the original teacher output and the super-source,

$$D_{\text{final}} = \mathrm{WBF}(D_{\text{V12}},\ S^*;\ w{=}(1.0,\ 0.25),\ \tau{=}0.70). \tag{5}$$

The blend weight w=0.25 on the super-source was selected by sweeping $w \in \{0.10, 0.15, 0.25, 0.35, 0.50\}$ on the meta-validation set only.

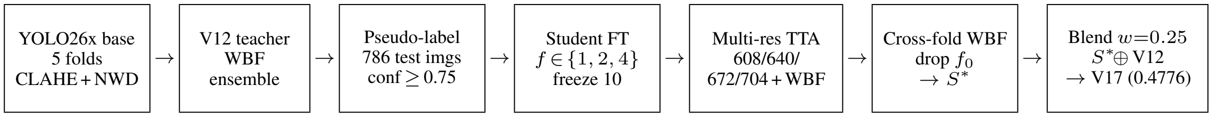

Fig. 1. Pipeline overview. A YOLO26x base detector (CLAHE input, pixel-space NWD loss) is trained per fold; the strongest pre-distillation ensemble (V12) pseudo-labels the 786 test images; per-fold students are fine-tuned on the fold-augmented data, run multi-resolution TTA fused by WBF, fused across folds (dropping the domain-overlapping fold 0) into a super-source S^*, and blended with V12 at w=0.25 to give the final submission V17.

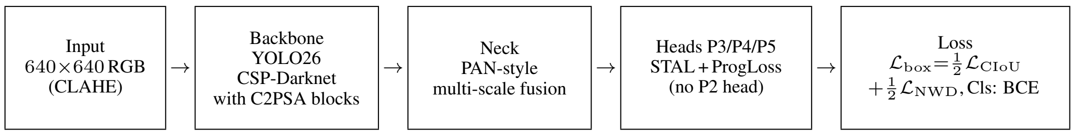

Fig. 2. YOLO26x base-detector architecture used in this work: a CSP-Darknet backbone with C2PSA attention blocks, a PAN-style multi-scale neck, and three detection heads at strides 8/16/32 (the optional P2 head is disabled, since YOLO26's built-in STAL and progressive-loss modules already address small targets). The box-regression loss blends CIoU with pixel-space Normalised Wasserstein Distance (NWD); classification uses binary cross-entropy.

4. EXPERIMENTS

4.1. Evaluation protocol

Metric. COCO mAP$_{50\text{-}95}$ computed with pycocotools [3] on the 401-image meta-validation set. Held-out test-set mAP is measured by the challenge platform on the undisclosed 786-image test partition: submissions are uploaded as a COCO-format JSON of detections, and the platform returns the official score (rate-limited to one submission per 12 h).

Role of cross-validation. The five-fold split serves an *ensemble-construction* role, not the conventional role of an out-of-fold generalisation estimator. Each fold detector becomes a member of the WBF teacher ensemble (Section 3.3); the per-fold *validation* mAPs are never used to estimate test performance or to select the final model. Cross-fold validation mAPs on the full training set are inflated ("leaked"), because every fold detector has, through the other four folds, seen the images held out by any single fold. We therefore measure generalisation with two separate sets: (i) the 401-image leak-isolated meta-validation set (Table 2), used only as a single-model gate, and (ii) the official held-out test set (Table 1), used for all model selection and final reporting.

Leak isolation. The 401-image meta-validation set is sampled exclusively from the fold-0 validation pool (seed 42, stratified by box count, small-object count and aspect ratio; all new models train on the remaining 2,753 images). It is leak-free only for models whose training partition does not overlap it: the cold fold-0 detector and the pseudo-FT students (trained on the meta-training pool) are clean, whereas the teacher ensemble V12, which includes folds 1–4 each trained on all of fold 0, is leaked on meta-validation. Meta-validation therefore cannot judge V12 or any blend that contains it (Section 4.2); the held-out test set is the only leak-free measurement for the full pipeline.

4.2. What transfers to genuinely unseen data

We measure how much each stage contributes on data no part of the pipeline has seen, using the official held-out test set (Table 1). This separates two quantities the meta-validation chain conflates: the strength of the student source in *isolation*, and its *marginal* value once added to the teacher.

Table 1. Contribution on the official held-out test set (the only leak-free instrument for the full pipeline). All three rows were scored by the challenge platform on the undisclosed 786-image partition.

Configuration	mAP$_{50\text{-}95}$	Δ
Plain 5-fold WBF ensemble	0.4720	—
Teacher V12 (+RF-DETR, CLAHE swap)	0.4755	+0.0035
Full pipeline V17 (+distill +TTA +drop-f_0 +blend)	**0.4776**	+0.0021

On unseen data, the full distillation, multi-resolution TTA, selective fold-dropping and blending stack adds only $+0.0021$ mAP$_{50\text{-}95}$ over submitting the teacher ensemble, and $+0.0056$ over a plain quality-weighted 5-fold WBF. The pipeline does not extend either: raising the blend weight to w=0.375 regressed to 0.4768 (-0.0008), and a second-generation pseudo-FT blend regressed to 0.4566 (-0.019, below the teacher). The marginal value of the super-source over the teacher is small, and positive only at a conservative blend weight.

4.3. Decomposing the student source (meta-validation)

The large meta-validation gains measure the construction of the student source in isolation, on the leak-isolated 401-image set (Table 2); every row is clean (the students train on the meta-training pool only). The chain is reproducible from the archived per-fold/per-resolution predictions (`scripts/verify_ablation.py`).

The AP chain $0.394 \to 0.434 \to 0.469 \to 0.471$ totals $+0.077$, with pseudo-distillation the single largest component (deltas $+0.040$, $+0.035$, $+0.002$). Small-box accuracy improves in step (AP$_S$ $0.356 \to 0.418$), while AP$_{75}$ stays the weakest column (0.522 on S^* versus 0.752 at AP$_{50}$), locating the residual error at high IoU, where sub-pixel stripe-height accuracy dominates. The contrast with Table 1 is the main result: the $+0.077$ buys a source that is, on unseen test, redundant with the teacher ensemble, which already encodes the same multi-fold knowledge. Adding S^* to V12 cannot improve meta-validation (V12 is leaked there, so its 0.515 ceiling is unreachable by a leak-free source) and improves held-out test by only $+0.0021$. The intermediate distillation stages were not measured on held-out test: each official submission was rate-limited to one per

Table 2. Construction of the student super-source S^* on the 401-image leak-isolated meta-validation set, with the size- and IoU-stratified breakdown (pycocotools). AP is mAP$_{50\text{-}95}$; AP$_S$ is the small-area average (COCO area $< 32^2$ px), the hardest RFI regime. These are single-source numbers; they do *not* include the (leaked) teacher.

Configuration	AP	AP$_{50}$	AP$_{75}$	AP$_S$
Cold fold-0 baseline	0.394	0.664	0.424	0.356
Pseudo-FT student, fold 4	0.434	0.706	0.464	0.372
+ Multi-res TTA (4 res, WBF)	0.469	0.746	0.511	0.411
+ 3-fold WBF (drop f_0) $= S^*$	**0.471**	**0.752**	**0.522**	**0.418**

Table 3. Cross-fold ensemble: effect of fold-0 inclusion.

Fold set	mAP$_{50\text{-}95}$
Fold 4 alone (TTA)	0.469
4-fold: F0+F1+F2+F4 equal weights	0.458
3-fold: F1+F2+F4 (drop F0)	**0.471**

12 h, so only the three ensemble end-points in Table 1 were scored on the true test partition.

4.4. Fold selection ablation

Table 3 shows that the 4-fold ensemble is *worse* than fold-4 alone, and the 3-fold (drop-F0) is best. This is explained by the domain overlap of fold-0 with the meta-validation pool: the fold-0 fine-tuned student produces correlated, low-diversity predictions that drag down the consensus.

4.5. Blend weight sweep

Table 4 reports the leaked fold-0 validation mAP$_{50\text{-}95}$ at three super-source blend weights together with the corresponding official held-out test scores. The V12 source is itself leaked on fold-0 val, so the baseline at $w{=}0$ is the leaked ceiling; adding the leak-free super-source decreases leaked val monotonically while improving held-out test up to $w{=}0.25$.

4.6. Negative results

We systematically evaluated several axes that did not transfer to the held-out test set.

RF-DETR ensemble. The teacher's RF-DETR-Large [17] component (fine-tuned from COCO, weight $1.5\times$) improves leaked fold-0 validation mAP by only $+0.001$ over the YOLO-only ensemble, and the official test set by $+0.0002$; across six checkpoint–weight variants the RF-DETR axis is saturated, so it is retained at minimal weight but contributes negligibly.

D-FINE-X. D-FINE-X [19] (Obj365→COCO pre-training) was fine-tuned on fold 0 for ten epochs. Final mAP$_{50\text{-}95}$ reached $3{\times}10^{-4}$, more than $1{,}000\times$ below the fold-0 baseline, despite a clean loss decrease from 60 to 30. DETR-family decoders appear to require many more epochs to adapt to the elongated aspect-ratio geometry of RFI boxes.

Score calibration. Temperature-scaled post-hoc calibration applied to the final blend yielded at most $+0.0002$ mAP, consistent with the teacher ensemble already encoding well-calibrated confidence scores.

Table 4. Blend weight sweep: V12 @ 1.0 + super-source @ w.

w	Leaked fold-0 val	Official test
0.00 (V12 alone)	0.5154	0.4755
0.25	0.5148	**0.4776**
0.375	0.5135	0.4768

Inference-only SAHI. Sliced-inference SAHI [20] at 320-pixel tiles regressed mAP by -0.085, consistent with tile-boundary fragmentation of long horizontal stripes.

5. DISCUSSION AND CONCLUSION

The cold per-fold baseline saturates around 0.394 mAP$_{50\text{-}95}$ (Table 2, top row): 2,753 images are too few to support reliable localisation of thin, low-contrast RFI stripes across the full IoU=0.50–0.95 sweep. Building the student source recovers $+0.077$ on meta-validation in three steps. *(i)* Pseudo-distillation from a multi-source ensemble teacher is the largest single component ($+0.040$): the soft signal in the teacher's confidence scores compensates for the small training set more effectively than the architectural alternatives we evaluated. *(ii)* Multi-resolution TTA between 608 and 704 px adds $+0.035$ because thin stripes a few pixels tall are resolved unevenly by YOLO's strided feature pyramid at any single input size. *(iii)* Dropping the domain-overlapping fold-0 from the cross-fold ensemble, a *smaller* ensemble, is net-positive because correlated predictions drag the WBF consensus when one source has seen the validation data during training.

This gain does not transfer. Almost all of the $+0.077$ is captured independently by the teacher ensemble: on the held-out test set (Table 1), adding the student super-source to the teacher gives only $+0.0021$, and a higher blend weight or a second pseudo-FT generation regresses below the teacher. The practical recommendation is therefore that a quality-weighted multi-fold ensemble recovers most of the attainable accuracy on a fresh SAR acquisition, and the distillation–TTA–blend stack adds little beyond it for its extra compute.

Several directions that improved leaked cross-fold validation did not transfer to held-out test: a second architectural family (RF-DETR) plateaued at $+0.0002$ across six checkpoint–weight variants; inference-only SAHI fragmented long horizontal stripes at tile boundaries (-0.085); and temperature-scaled calibration could not improve a teacher whose confidences were already well-conditioned by WBF fusion. Horizontal-flip TTA cost -6.5% relative, consistent with SAR side-looking geometry being asymmetric in azimuth–range [21]. The largest remaining gap is at high IoU thresholds, where sub-pixel height accuracy dominates and a 640-px input under-resolves the median $\approx$10 px stripe; ranked by expected return, the most promising next moves are (a) sliced *training* at 320-px tiles so the backbone sees RFI at doubled relative height, (b) a DETR-family detector with a different decoder trained for substantially more epochs than the ten that proved insufficient for D-FINE-X [22], and (c) a second pseudo-label refresh with V17 itself as the new teacher.

The final pipeline — YOLO26x with pixel-space NWD regression loss, ensemble-teacher pseudo-distillation, multi-resolution TTA and a selective cross-fold WBF — reaches 0.4776 mAP$_{50\text{-}95}$ on the held-out test set; on meta-validation the ablation chain in Table 2 totals $+0.077$ over the per-fold baseline. The submission placed the *UdeM AI* team 28th of 172 teams on the public leaderboard of ESA ClearSAR Track 1. All training and fine-tuning were performed on a single cloud NVIDIA A100 GPU.

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